Tutorial 1

Online editor and compiler

Tutorial Structure

- Deadlines and Instructions
- Plagiarism
- Implementation:
    - Today's program
    - Some things we need to know
- Testing our example code
- Submission

1. Deadline and Instructions

- Practicals typically run for a week
- No extension will be granted - **start early**
- Read instructions and ask tutors for help
    - If you do your practical during the practical session tutors will be there to answer questions as they come up
- Do not make assumptions, if you do not understand - **ask!!!**

2. Plagiarism

- Disciplinary action will be taken against students who commit plagiarism. Plagiarism includes:
  - copying someone else’s work without consent,
  - copying a friend’s work (even with permission) and
  - copying material from the Internet.
- Copying will not be tolerated in this course
- Fitchfork (the platform where you submit your practicals) can detect plagiarism

3. Programming Environment and Structure

1. Online editors and compiler

- C++98 version
  - https://cpp.sh/
  - Others:
    - **https://www.onlinegdb.com/online_c++_compiler**
    - https://www.programiz.com/cpp-programming/online-compiler/
    - https://www.w3schools.com/cpp/trycpp.asp?filename=demo_compiler

2. Basic Program Structure and syntax
   - Headers
   - Main function
     - Curly brackets
     - Return

4. Today’s Program

- Use example output as reference
- **Formatting is important!** (you get auto marked so you need to match the expected output exactly)
  - Decimal vs whole numbers
  - Case sensitive
  - Match whole phrase/statement including spaces
- Different scenarios (we won’t only test with the given examples)
  - Test with different user input (once user input has been taught)
- Example case study:
  - Develop a program that will convert the users current balance to foreign currency:
  - Example output:

```
Welcome to the foreign exchange bank!

How much would you like to convert  R1

R1 = $19.07
```

5. Some things we need:

* If this is still scary today, don't worry. You will be taught in detail what everything means during lectures. We just need some building blocks to run code in the online editor. For today, just copy exactly what's on the slides.

iostream	A standard library (this is code that has been written already and we can just use it)
Main function	This is where we will put all our code for now
std::cout	This is how we print
<<	This is like a "pipe" that takes the stuff from the left and puts it into the cout to be printed
" "	When we want to print characters we need to put it in " ", because we want to print it directly without treating it as pieces of code. Almost like a direct quote
std::endl	This just ends the current line you're printing (like pressing Enter)
int mybalance	This is where we will store the current value. Think of it as a box (labelled "mybalance") that can store integers (numbers without decimals)
*	Multiplication (x) in C++

6. Testing our example code

```cpp
#include <iostream>
int main(){
    std::cout << "Welcome to the foreign exchange bank!" << std::endl;
    int mybalance = 1;
    std::cout << "R" << mybalance << " = $" << mybalance * 19.07 << std::endl;
    return 0;
}
```

7. Submission

- Copying/Downloading from online editor
- Save to cpp file
- Compressing
  - Windows 10:
    - Select the file, then click on "Send to", then "Compressed (zipped) folder" to compress it
  - Windows 11:
    - Select file(s), right-click, then “Compress to ZIP file”
  - Mac:
    - Select file(s), right-click, then “Compress” (will include file name if only 1 file)
  - Ubuntu:
    - Open a terminal (you can right click and open one in the directory where the files you want to zip are)
    - Navigate to the right directory (if you aren’t already there) using cd
    - `zip name of zip <files you_want_to_zip>`
    - `Eg. zip myTask main.cpp`
- Be sure to rename your compressed archive according to the specification

Any questions?